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High temperature oxidation resistance of W-containing high entropy alloys

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ABSTRACT

Tungsten containing refractory high entropy alloys (HEAs) are potential candidate for high temperature applications owing to their remarkable mechanical properties. However, the oxidation resistance of W containing HEAs at high temperatures is problematic, and it received less attention. In this work, several equiatomic W containing WTaTiCr, WMoTaTiCr, WTaAlCr, and WMoTaAlCr refectory HEAs were fabricated by arc melting technique and their oxidation behavior at 1000 °C for 10 h was compared with MoTaTiCr medium entropy alloy (MEA). The as cast microstructure of the MoTaTiCr, WTaTiCr, WMoTaTiCr, and WMoTaAlCr alloys exhibited a single BCC solid solution, while the as cast WTaAlCr MEA showed some extent of intermetallic compounds along with the BCC phase. The best oxidation resistance was obtained for MoTaTiCr MEA, which was protected by a continuous $CrTaO_4$ scale layer. The formation of tungsten oxides (mainly WO_3) prohibited the evolution of a protective oxide layer of $AlTaO_4$, $CrTaO_4$ or Al_2O_3 on the surface of W containing alloys. WMoTaAlCr HEA exhibited the best oxidation resistance among all the W containing alloys, which was protected by a nearly continuous rutile type $(Al,Cr)TaO_4$ oxide layer.

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1. Introduction

High entropy alloys (HEAs) are defined as alloying system consisted of five or more principal elements with typically equiatomic or near equiatomic ratio, which occasionally exhibit extraordinary properties [1,2]. Among different types of HEAs, refractory HEAs (RHEAs) are considered as the most promising alloy systems to design new materials for high temperature applications beyond Ni based superalloys [3,4]. RHEAs were first introduced by Senkov et al. [5] where they fabricated WNbMoTa and WNbMoTaV alloys for high-temperature load-bearing applications. Afterward, several RHEAs with outstanding mechanical properties have been developed [6]. While most researches have been focused on the mechanical properties of RHEAs at high temperature, several works have recently considered the high temperature oxidation behavior of RHEAs [7–10].

Oxidation resistance is a crucial property for materials working for long time at high temperatures since it can severely deteriorate mechanical properties of the materials. Al, Si and Cr are usually considered as the protective oxide forming elements (e.g., Al_2O_3, SiO_2, Cr_2O_3) to prevent materials from further oxidation [11]. In contrast to conventional alloys, in which only a small amount of the stable- and dense-oxide forming elements could be added to improve the oxidation resistance, HEAs provide the possibility of incorporating higher amounts of these elements without sacrificing their mechanical properties. Recently, promising oxidation resistance has been found in (Mo-Ta-Ti-Nb-Zr-Hf)-(Cr, Al, Si) systems in the temperature range of 900–1100 °C owing to the formation of a protective complex $CrTaO_4$ oxide layer [12–15]. It was revealed that the dense $CrTaO_4$ layer could prohibit the outward diffusion of the constituent elements and provide outstanding oxidation resistance, a similar mechanism which has been already observed in Ni based superalloys [16,17]. The equiatomic TaMoTiCrAl HEA, which developed by Gorr et all. [12], turned out to be one of the most oxidation resistant HEA reported to date. Recently, Li et al. [14] found out that an even more oxidation resistance MoTaTiCr medium entropy alloy (MEA) could be obtained when Al is removed from the quinary Ta-

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Mo-Ti-Cr-Al system. These recent studies pave the pathway for development MEAs/HEAs with outstanding oxidation resistance for long term application at high temperatures.

Among RHEAs, W-containing MEAs/HEAs are potential candidate for high temperature applications such as plasma facing material in nuclear reactors owing to high melting temperature, high thermal conductivity and small tritium retention of tungsten [18,19]. However, a systematic study on the oxidation resistance of W-containing HEAs is still lacking. Therefore, in this study, we investigate the oxidation resistance of several W-containing refractory MEAs/HEAs at high temperatures to further examine the formation of a dense complex oxides layer on the surface of these alloys. Here, MoTaTiCr MEA which reported to exhibit excellent resistance against high temperature oxidation [14], was adapted as the prototype system to develop W containing alloys with superior high temperature oxidation resistance. By tailoring the chemical composition of MoTaTiCr MEA, five distinct compositions of WMoTaTiCr (by adding W to MoTaTiCr), WTaTiCr (to study the effect of Mo removal), WTaAlCr, WMoTaAlCr (to utilize the beneficial effect of Al_2O_3), and WMoVTiCr (to study the effect of V) were developed. The high temperature oxidation of these alloys were thoroughly analyzed at 1000 °C for 10 h.

2. Material and methods

Equimolar WTaTiCr, WMoTaTiCr, WTaAlCr, WMoTaAlCr, and WMoVTiCr alloys were fabricated by arc-melting process under argon atmosphere using high purity (> 99.9 wt%) metal powders. Pure powders were first weighted and thoroughly mixed in an agate mortar. The mixed powders were then loaded into a water-cooled copper mold and arc melted under Ar atmosphere. The bottoms were flipped over and remelted five times to ensure chemical homogeneity. The fabricated bottoms were then cut and polished for structural analysis. X-ray diffraction (XRD) was collected from the top surface of the as-cast ingots by a Rigaku Ultima IV X-ray diffractometer using Cu–Kα radiation (λ = 0.154 06 Å) to reveal crystal structures of the samples. Microstructural analysis was studied on a scanning electron microscope (SEM) Jeol JSM7001F equipped with energy-dispersive X-ray spectroscopy detector (EDS; Oxford INCA X-max 80). Samples with a dimension of 5 mm × 5 mm × 3 mm were cut from the ingots and polished for oxidation tests. Quantitative oxidation resistance of the samples were measured using a Q-1500D derivatograph (Paulik and Erdei, Hungary) under isothermal conditions at 1000 °C for 10 h in air atmosphere. To reveal the oxidation behavior of the materials, the oxidized surfaces were then thoroughly analyzed by XRD and SEM-EDS analysis. Moreover, the microstructure and oxidation resistance of the samples were compared with MoTaTiCr MEA fabricated by a same arc melting process.

3. Results and discussion

3.1. Microstructural analysis of as cast alloys

The XRD spectra of the as cast samples are shown in Fig. 1. A nearly single phase BCC solid solution was obtained for all the samples except WTaAlCr, which exhibited the presence of intermetallic compounds (IMCs) such as $TaAl_3$ and Al_8Cr_5. This may be unexpected as the atomic mismatch (δ), mixing entropy (ΔS_{mix}), and mixing enthalpy (ΔH_{mix}) (Table 1) all fall within the established range for the formation of solid solution phases [20,21]. However, WTaAlCr exhibited the most negative value of mixing enthalpy (−10.98 kJ mol^{-1}) among all the samples, which is near to the values reported for a mixture of a solid solution and an intermetallic compound (ΔH_{mix} ≈ 10 kJ mol^{-1}, δ ≈ 6–7%) [22]. It is well known that the tendency of the elements to randomly distribute in the solution is lower at more negative ΔH_{mix} or more positive ΔH_{mix} due to higher binding energy between the elements or their miscibility gaps, respectively. Besides, there exist high differences between Al and refractory elements (W and Ta) in chemical and physical properties such as melting point, valence electron concentration, and electronegativity, which are typical conditions for the formation of IMCs. When Mo is added to WTaAlCr, the atomic mismatch decreases, the entropy of mixing increases, and the enthalpy of mixing become less negative, all these favor the formation of a single solid solution phase. It should be noted that these structural and thermodynamic parameters must be considered as a rough predictor not a general rule for the formation of solid solution, as already observed for several HEAs [23]. With the addition of V and Al, the diffraction peaks shifted towards higher angles, indicating a decrease in the lattice parameter due to the smaller size of Al and V compared to the Ti and Ta, respectively. It should be also noted that minor oxide phases were also detected only on the top surface of the as-cast samples. The internal part of ingots were free of oxides, as no oxide particles could be detected in the SEM/EDS analysis of the cross-section of the alloys (see Fig. 2). This is due to the low density of oxides and surface tension of liquid alloy, which expulsed all the oxide particles (or droplets) to the surface of the alloys.

The back scattered-electron SEM (BSE-SEM) micrographs of all the as-cast refractory HEAs are shown in Fig. 2. It can be observed that all the as cast samples exhibit dendrite microstructure with light gray and dark gray contrasts being dendritic and inter-dendritic regions, respectively. An explicit elemental segregation could be also detected for all the samples. Generally, the higher melting point elements such as W, Mo and Ta segregated into the dendritic arms, while the elements with lower melting point such as Al, Cr and Ti segregated into the inter-dendritic regions, as can be seen in the EDS maps. However, two anomalies from this trend were also observed. Ta exhibited a nearly uniform distribution in the Al containing WTaAlCr (Fig. 2d) and WMoTaAlCr (Fig. 2e) alloys, and Mo segregated partially into inter-dendritic regions for the alloys containing both Mo and W elements (Fig. 2b, e and f). Segregations in HEAs may be inferred from the melting point of the individual elements, and the mutual solubilities and enthalpies of mixing of the binary element pairs. Among all the binary element pairs in W-Mo-Ta-Al-Cr system, Ta-Al pair exhibits the highest negative enthalpy of mixing (−19 kJ mol^{-1}) [24], which may be the main reason for more uniform distribution of Ta in the dendritic (melting point induced segregation) and inter-dendritic (Al induced segregation) regions upon adding Al to the refractory HEAs. The more segregation of Mo into inter-dendritic regions in W-containing alloys is unexpected when considering the mutual solubility of Mo in W in the entire temperature ranges below solidus temperature [25], and their nearly ideal enthalpy of mixing (ΔH_{mix} = 0) [24]. A same segregation behavior for Mo has been also observed in the as cast WMoTiAlCr HEA [26]. This may be partially ascribed to the more negative enthalpy of mixing of Mo-Ti (−4 kJ mol^{-1}) and Mo-Al (−5.1 kJ mol^{-1}) binary pairs compared to the Mo-W pair.

The chemical composition of the dendritic and interdendritic regions, and the overall composition of the alloys are shown in Table 2. Generally, all the as cast alloys exhibited some degree of Cr- or Al-deficiency due to the higher evaporation of these elements (compared to the refractory elements) during arc melting process. In spite of this elemental segregations, XRD spectra revealed a single BCC structure for all the samples except WTaAlCr, as discussed above. The high magnification BSE-SEM image of WTaAlCr that is shown in Supplementary information Fig. S1, clearly indicates the

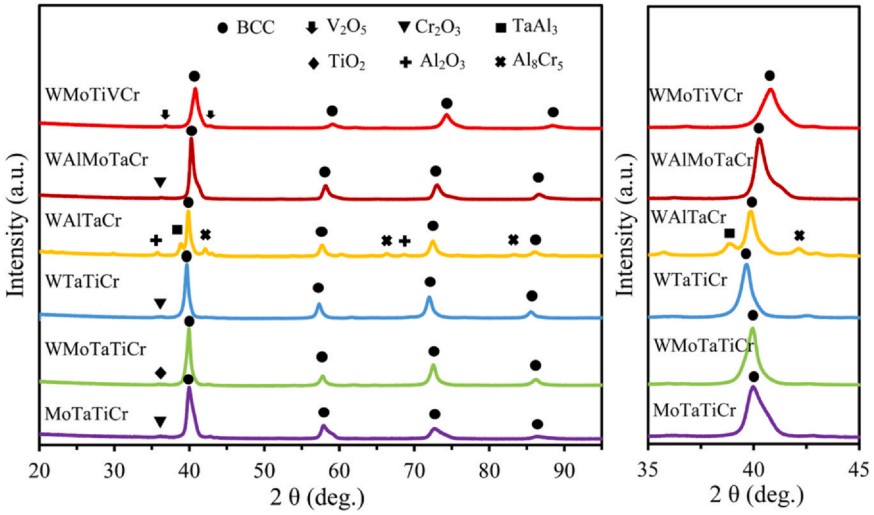


Fig. 1. (a) XRD patterns of the as-cast RHEAs, and (b) an enlarged image of the main diffraction peaks showing the bragg peaks shift.

presence of a new phase with different contrast to those of dendritic and interdendritic regions. Moreover, some cracks are also present in the microstructure of this alloy, which likely occurred during rapid cooling regime of arc melting due to the presence of brittle intermetallic compounds. For other samples, it may be assumed that both dendritic and interdendritic regions have a same BCC structure with different chemical compositions but a nearly same lattice parameters. This is a common feature between as cast refractory HEAs, which has been already observed in other alloys [27].

3.2. Oxidation behavior

Dynamic oxidation weight gains were measured on a derivatograph device under isothermal conditions at 1000 °C for 10 h in air atmosphere. The results are shown in Fig. 3. The data for WMoTiCrV are not presented in Fig. 3 because this sample exhibited a dramatically high mass change after 10 h oxidation at 1000 °C. As expected, MoTaTiCr exhibited the best oxidation resistance among all the samples. A relatively small mass gain of about 2.1 mg cm^{-2} was obtained for MoTaTiCr after 10 h oxidation at 1000 °C. This value is in consistent with the previously reported value of 0.93 mg cm^{-2} for 5 h oxidation at 1000 °C [14]. By substituting W for Mo in MoTaTiCr, i.e., WTaTiCr, the mass gain significantly increased upon isothermal exposure, and a mass gain of about 9.3 mg cm^{-2} was recorded for this sample after 10 h oxidation. Addition of Mo to WTaTiCr, i.e., WMoTaTiCr, further deteriorated the oxidation resistance of WTaTiCr MEA, and the mass change after 10 h was about 14.49 mg cm^{-2}, significantly higher than its MEA counterpart. We considered Al as a potential candidate to further increase the oxidation resistance of W-containing alloys. However, substituting Al for Ti in WTaTiCr, significantly increased the mass gain. Indeed, as can be seen in Fig. 3, WTaAlCr showed the lowest oxidation resistance with an overall mass gain of 17.55 mg cm^{-2}. Finally, by adding Mo to WTaAlCr, we developed WMoTaAlCr HEA which presented superior oxidation resistance compared to its MEA counterpart, but still higher mass gain than MoTaTiCr MEA. The mass gain of WMoTaAlCr HEA after 10 h oxidation at 1000 °C was about 7.2 mg cm^{-2}, comparable to that reported for WMoTiAlCr (~5.5 mg cm^{-2}) [26] and significantly smaller than that reported for Nb-Cr-V-W-Ta (~ 200 mg cm^{-2}) [28] HEAs. It should be also noted that the recorded mass gain for the alloys is affected by the level of micro-porosities or micro-cracks in the samples. Especially, we believe that the low oxidation resistance of WTaAlCr MEA is mainly due to the presence of microcracks not its intrinsic low oxidation resistance. However, further works are required for WTaAlCr MEA to accurately judge about the effect of intermetallic compounds and the ensuing cracking on its oxidation behavior. Nevertheless, WMoTaAlCr HEA is among the best oxidation resistance W containing HEAs reported so far.

The surface morphology of the oxide layers is shown in Fig. 4. The corresponding EDS elemental maps are also shown in the Supplementary information Fig. S2. A dense and continuous scale layer covered most area of the oxidized surface of MoTaTiCr MEA (Fig. 4a). However, some porous spherical crystals could be also observed on the surface of MoTaTiCr. As can be seen in Fig. S2, these spherical crystals are mainly Ti-rich oxide phases. A highly porous scale layer consisted of large voids and cracks formed on the surface of WMoTaTiCr, WTaTiCr, and WTaAlCr alloys. Especially, WTaAlCr MEA exhibited a completely disintegrated oxide layer with high degree of spallation. Explicit traces of spallation could be also detected on the surface of WMoTaTiCr and WTaTiCr alloys. This porous oxide layer is consistent with their low resistance against high temperature oxidation. Among all the W containing alloys, WMoTaAlCr exhibited the most continuous scale layer with significantly lower porosities and cracks. Quantitative EDS analysis performed on

Table 1
The structural (atomic mismatch, δ) and thermodynamic (mixing enthalpy, ΔH_{mix}, and mixing entropy, ΔS_{mix}) parameters for the investigated alloys.

	MoTaTiCr	WMoTaTiCr	WTaTiCr	WTaAlCr	WMoTaAlCr	WMoTiVCr
δ	5.93	5.32	5.91	5.43	4.86	5.16
ΔS_{mix}(J K^{-1} mol^{-1})	11.53	13.38	11.53	11.53	13.38	13.38
ΔH_{mix}(kJ mol^{-1})	-5.5	-5.44	-6.25	-10.98	-8.64	-3.36

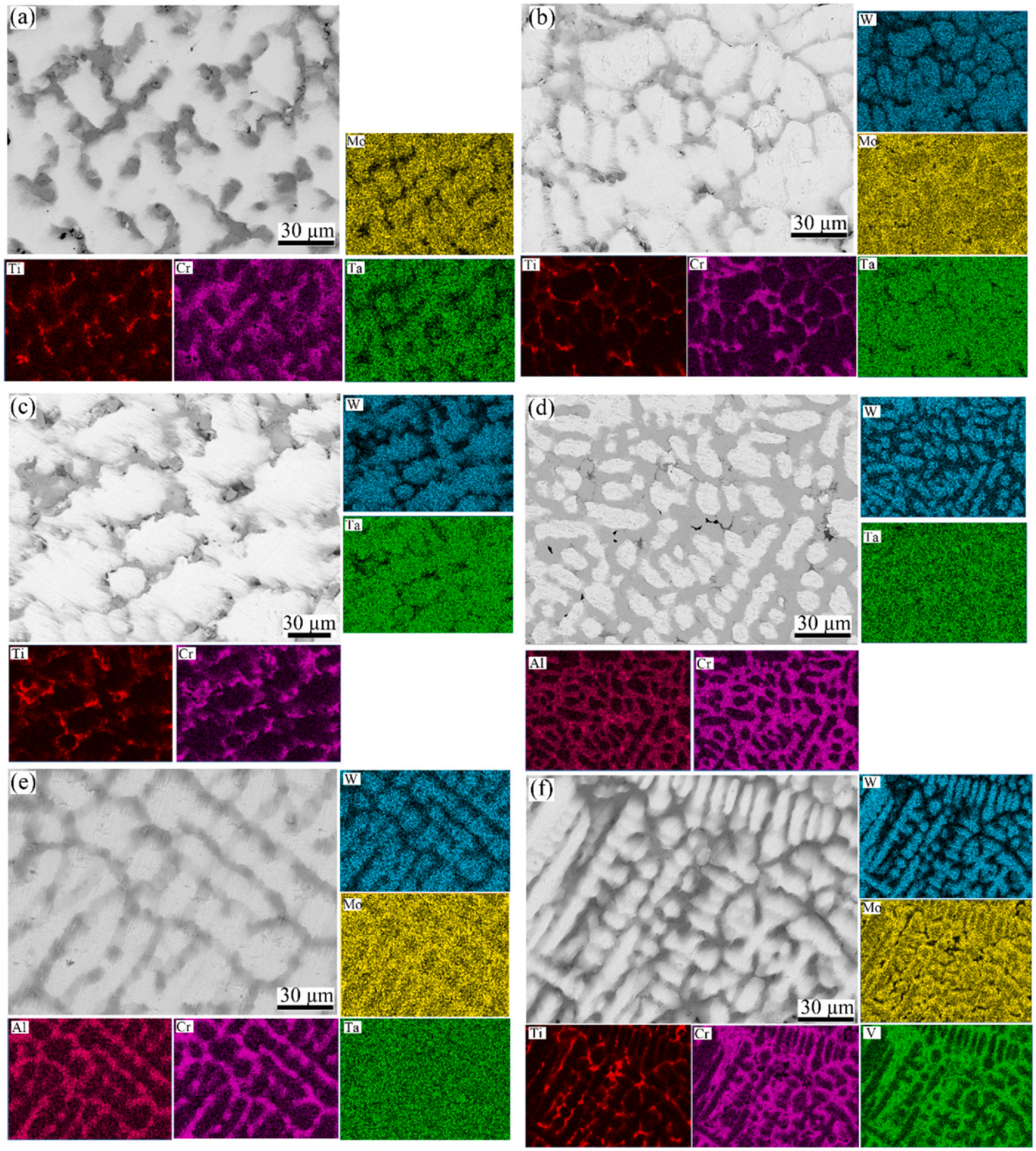


Fig. 2. BSE-SEM micrographs and the corresponding EDS elemental maps of the constituent elements for (a) MoTaTiCr, (b) WMoTaTiCr, (c) WTaTiCr, (d) WTaAlCr, (e) WMoTaAlCr, and (f) WMoTiCrV HEAs.

the surface of scale layers (Table S1) indicated the presence of all constituent elements, except Mo, on the surface of the samples. This points out the likely evaporation of Mo oxide, since MoO_3 is reported to be highly volatile above 800 °C [29–31].

Fig. 5 shows the XRD spectra collected from the oxidized surface of the samples. The oxide layer on the surface of MoTaTiCr is mainly consisted of rutile type $CrTaO_4$, in consistent with the previous results [14]. Addition of W to MoTaTiCr MEA highly changed the

Table 2
EDS chemical compositions measured from the dendritic and interdendritic regions of as-cast HEAs. The average values of 5 measurements and the corresponding standard errors are reported.

Alloys	EDS chemical composition (at%)	W	Mo	Ta	Ti	Al	Cr	V
MoTaTiCr	Dendrite (D)	–	26.59	29.05	21.81	–	22.56	–
	Interdendrite region (ID)	–	8.50	11.70	39.03	–	40.77	–
	Average composition	–	25.07	26.81	24.40	–	23.72	–
WMoTaTiCr	D	26.57	22.70	25.09	14.07		11.57	
	ID	8.28	17.14	16.17	30.35	–	28.06	–
	Average composition	22.42	21.44	23.07	17.76	–	15.31	–
WTaTiCr	D	34.40	–	32.76	19.62	–	13.21	
	ID	9.11	–	19.42	35.31	–	36.17	–
	Average composition	28.34	–	29.56	23.38	–	18.71	–
WTaAlCr	D	31.41	–	29.79	–	16.27	22.53	–
	ID	9.23		24.38	–	28.62	37.76	–
	Average composition	28.00	–	28.92	–	18.11	24.97	–
WMoTaAlCr	D	34.19	22.61	23.16	–	9.74	10.30	–
	ID	5.63	16.33	17.31	–	22.67	38.06	–
	Average composition	27.05	21.04	21.70	–	12.97	17.24	–
WMoTiCrV	D	28.81	24.04	–	13.60	–	13.69	19.86
	ID	6.72	13.52	–	31.04	–	25.53	23.18
	Average composition	20.29	20.11	–	21.24	–	17.95	20.40

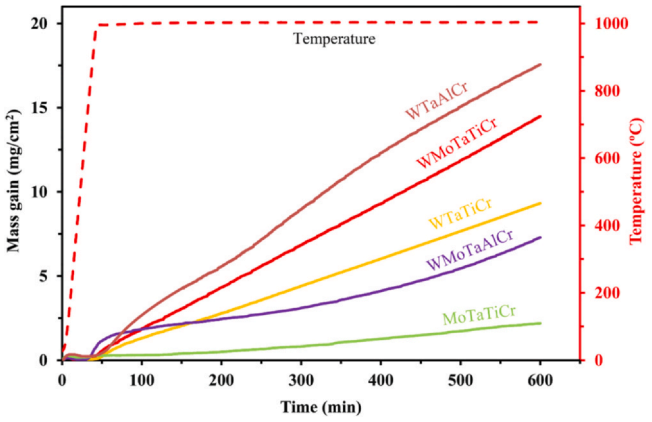


Fig. 3. Dynamic oxidation weight gain of the samples during 10 h oxidation at 1000 °C in air atmosphere.

composition of the oxide layer. While the diffraction peaks of rutile type structures ($CrTaO_4$ or TiO_2) could be still observed in the XRD spectra, a considerable amount of tungsten oxides such as WO_3, W_2O_5 and $Ta_{15}Wo_{40.5}$ formed on the surface of WMoTaTiCr HEA after oxidation at 1000 °C. A nearly same oxide composition was also observed for WTaTiCr MEA. The rutile type structures ($CrTaO_4$ and $AlTaO_4$) was also formed on the oxidized surface of WTaAlCr and WMoTaAlCr alloys. The main difference between these two Al containing alloys was the presence of a high intensity diffraction peak of WO_3 in WTaAlCr MEA, which was absent in WMoTaAlCr HEA. The oxidized surface of WMoTiVCr was also characterized by rutile type oxide, WO_3 and different vanadium oxides. Interestingly, no trace of Mo oxides could be detected on the surface of the alloys. It is clear that the binary or ternary oxides of W, Ti, and V cannot form a dense continuous layer on the surface to hinder oxygen diffusion and provide resistance against high temperature oxidation.

The cross section of the oxide layers was further analyzed by using BSE-SEM and EDS elemental map to study the nature of the oxide phases formed on the surface of the alloys. The results are shown in Fig. 6. For MoTaTiCr MEA, a relatively thin and dense oxide layer consisting of mainly Ta and Cr oxides ($CrTaO_4$ phase as determined by XRD spectra) formed on the surface of the alloy (Fig. 6a). This is consistent with the previously observed dense $CrTaO_4$ oxide layer on the surface of MoTaTiCr MEA. On the other hand, the oxide layers formed on the surface of W containing alloys exhibited a completely different morphology. The light gray dendritic arms and the dark gray interdendritic regions could be clearly detected in the oxide layers. The bright contrasts are likely W and Ta rich oxides, while the gray contrast regions are rich in Cr, Ti and Al depending on the chemical composition of the alloys. The thickest oxide layer (about 350 µm) formed on the surface of WMoTaTiCr HEA (Fig. 6b). By removing Mo from WMoTaTiCr, the oxide layer significantly decrease to about 200 µm for WTaTiCr MEA (Fig. 6c). EDS analysis indicated that the oxide layers are rich in W, Ta and Cr, and poor in Ti for both WMoTaTiCr and WTaTiCr alloys (Table 3). Considering the XRD and EDS analysis, it can be concluded that the oxide layers on these alloys are mainly consisted of WO_3 and $CrTaO_4$ phases. It seems that addition of Al could further decrease the thickness of oxide layer on the surface of refractory WTaTiCr and WMoTaTiCr alloys. However, it should be noted that WTaAlCr MEA exhibited high degree of spallation during oxidation test at 1000 °C (Fig. 6.d), and the thickness of visualized layer in Fig. 6d is significantly lower than the real thickness of the oxidized material. Among all the W containing alloys, WMoTaAlCr showed the most compacted scale layer with lower thickness and porosities (Fig. 6e). The EDS analysis also indicated the presence of all W, Ta, Cr and Al elements in the oxide layer. However, based on the XRD and EDS analysis, it cannot be precisely judged that the rutile type oxides in the scale layers of these alloys are $AlTaO_4$, $CrTaO_4$, a mixture of both phases, or a multicomponent rutile type (Al,Cr)TaO_4 oxide phase. Besides, the dilution of Mo in the scale layers is obvious for the Mo containing alloys. This can be ascribed to the high volatile behavior of Mo oxides at temperatures above 800 °C [31]. Moreover, evaporation of MoO_3 likely leaves porosities in the scale layer, which acts as oxygen diffusion path and deteriorate the oxidation resistance of the alloys.

3.3. Discussion

It has been already reported that the formation of a continuous $CrTaO_4$ oxide layer on the surface of HEAs can effectively retard oxygen diffusion and provide great protection against oxidation at

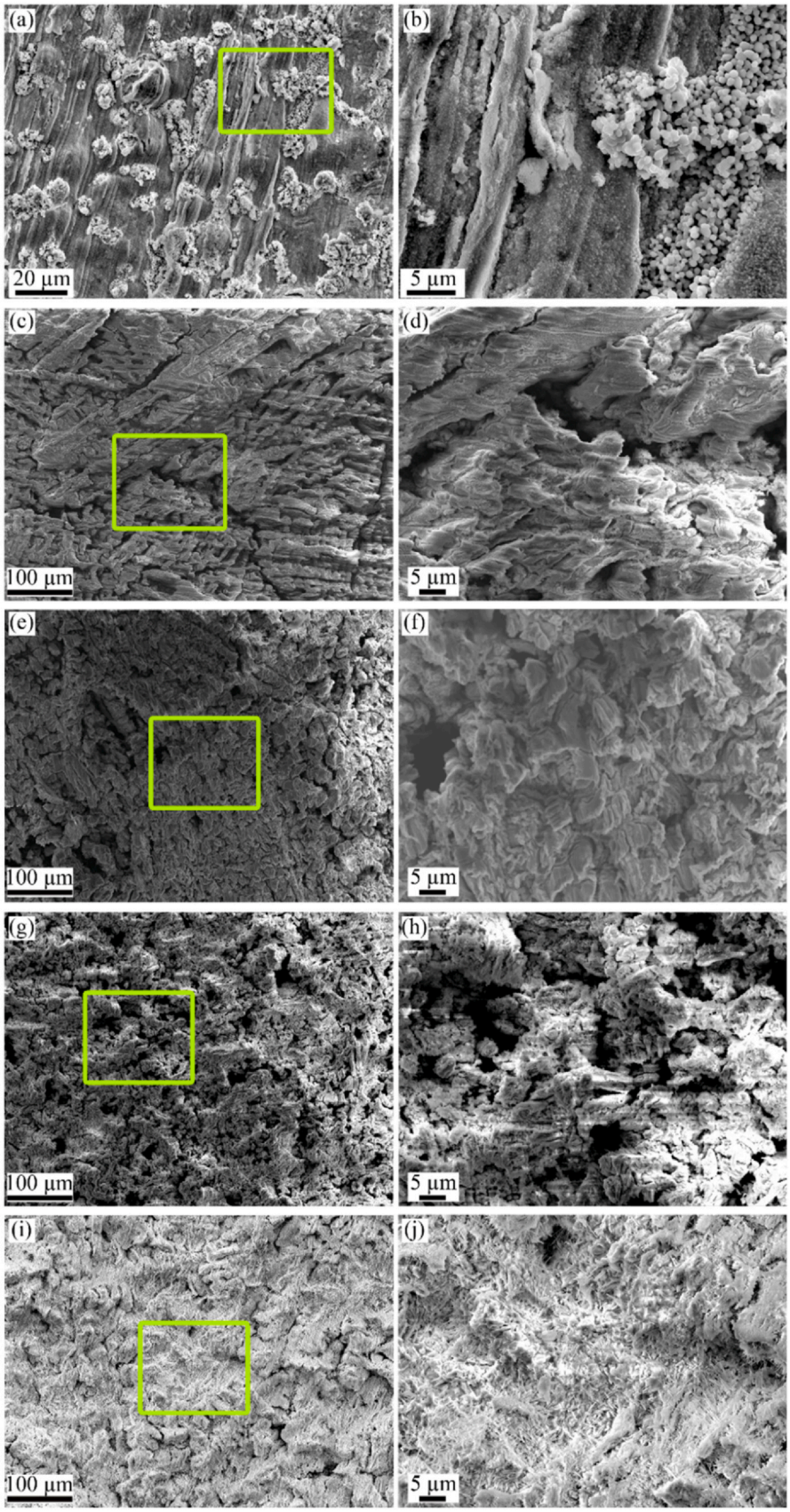


Fig. 4. SEM micrographs of the surface oxide layer formed after 10 h oxidation at 1000 °C, a high magnification of the indicated area is also shown: (a, b) MoTaTiCr, (c, d) WMoTaTiCr, (e, f) WTaTiCr, (g, h) WTaAlCr, (i, j) WMoTaAlCr HEAs.

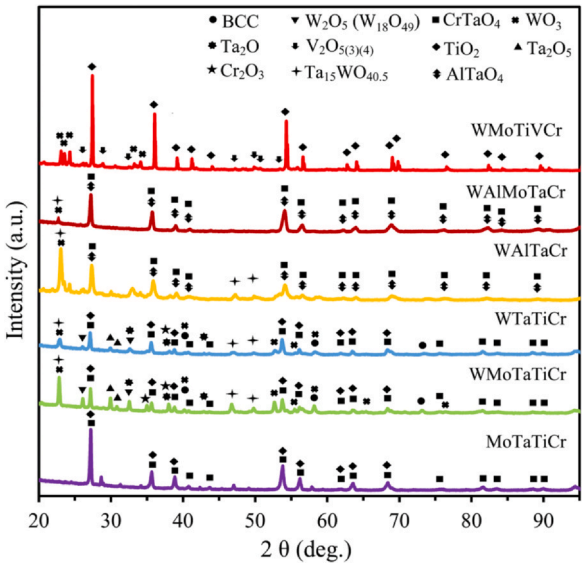


Fig. 5. XRD spectra of the samples after 10 h oxidation at 1000 °C.

high temperatures [12–14]. In this study, a nearly same protective $CrTaO_4$ oxide layer was observed for MoTaTiCr MEA. However, the thickness of $CrTaO_4$ rich oxide layer formed on the surface of MoTaTiCr MEA in our study was higher than that observed by Li et al. [14]. This discrepancy may be ascribed to the different degree of initial micro-cracks present in the samples. Moreover, a clear Mo removal from the oxide layers was detected for all the samples owing to the volatile nature of MoO_3 and its evaporation at high temperatures [8,32]. Therefore, it may be expected that underestimated mass gains were recorded during quantitative oxidation test.

Typically, the oxidation behavior of metallic alloys are governed by two extreme situations. The first case is where the alloys cannot develop a continuous protective scale layer because of either the formation of cracks or the evaporation of the constituent elements of the oxide layer. Cracks in the oxide layer may form due to the volume mismatch between the oxide layer and the alloy or between the different phases that are present in the oxide layer. Moreover, the cracks and micro-porosities occurred during manufacturing process also hinder the formation of a protective scale layer. Any discontinuity in the scale layer provides diffusion path for oxygen atoms and leads to catastrophic oxidation of the beneath alloy. In the second situation, the alloy develops a dense and continuous oxide layer on its surface, which retards oxygen diffusion and prevent further oxidation of the alloy. Here, the significantly lower oxidation resistance of WTaAlCr MEA compared to other alloys may be attributed to the internal cracking of this alloy during rapid cooling regime of arc melting process due to the presence of brittle intermetallic compounds (see Fig. S1). The oxidation behavior of other alloys could be inferred from the composition of the surface oxide layer. According to the Ellingham diagram, the formation of all observed oxide phases ($CrTaO_4$, (Al, Cr) TaO_4), TiO_2, WO_3, etc.) are thermodynamically favorable at 1000 °C, as the changes in the Gibbs free energy of the formation of all oxides are relatively large negative values [33–35]. However, it seems that the kinetic of these oxidation reactions are affected by the composition of the alloys. Especially, by comparing the oxidation behavior of WMoTaAlCr and WMoTaTiCr, it seems that Al has a favorable effect on the kinetic of (Al, Cr)TaO_4 formation reaction, which results in a relatively continuous oxide layer and superior oxidation resistance in WMoTaAlCr HEA. Moreover, by removing Mo from WMoTaTiCr, the oxidation resistance further increases, likely due to the evaporation of Mo oxide at high temperatures, which leaves some degree of porosity in the oxide layer. Therefore, WTaTiAlCr HEA would be a proper composition to develop oxidation resistant HEAs for future works, if the alloy shows a single high entropy phase without significant IMCs.

In RHEAs, the sluggish diffusion of atoms may limit the outward migration of elements even at high temperatures, and therefore, the inward diffusion of oxygen would play a more significant role during the oxidation regime. As observed in Fig. 6, no considerable depletion zone of the elements (except Mo which is evaporated) could be resolved in the oxidized microstructures, illustrating limited outward diffusion of the constituent elements in all the alloys. Moreover, since the microstructure of the as cast alloys was composed of W- and Ta-rich dendritic arms and Ti-, Cr- and Al-rich interdendritic regions, the inadequate diffusion kinetic would result in heterogeneous oxidation of these compositionally different areas. That is, W-rich region in dendritic arms mainly formed WO_3, while Cr-rich and Al-rich region in interdendritic area could form $CrTaO_4$ and $AlTaO_4$, respectively. This heterogeneous oxidation is accompanied with different volume changes, and results in volume defects within the scale layer. In this case, the scale layer could not act as an effective diffusion barrier layer. The oxidation induced volume change may be quantified by using Pilling-Bedworth ratio (PBR) [14,36], which is defined as the ratio of the oxide volume to the volume of the corresponding alloy ($PRB=V_{oxide}/V_{alloy}$). Only when the PBR falls in the range of 1–2, a continuous dense oxide layer may be formed and protect the alloy from further oxidation [37,38]. When PBR is either smaller than 1 or larger than 2 the oxide scale cannot be continuous and protective due to the small volume of oxides or internal stress induced cracking and falling off, respectively [37,38].

To shed light on the oxidation behavior of the alloys, we estimated the PBR values for different oxide phases formed on the surface of the alloys by using XRD analysis. For simplicity, we considered an identical volume for each atom occupying the BCC lattice cites of the MEAs/HEAs. In BCC lattice, each unit cell with a volume of a^3 (a is the lattice parameter) contains two atoms, and the PBR may be written as [36]: $PBR = V_{oxide}/0.5na^3$, where V_{oxide} and n are the volume and the number of the metallic atoms in the oxide unit cell. The calculated PBR values are shown in Table 4. Evidently, WO_3 and $Ta_{15}WO_{40.5}$ exhibit a relatively large PBR value to form a continuous protective oxide layer. This explains why W containing alloys could be easily oxidized. On the other hand, oxides such as $CrTaO_4$, $AlTaO_4$, Al_2O_3, and even TiO_2 show ideal PBR for protective oxide layers. However, TiO_2 usually does not form a continuous layer to provide high temperature protection [8]. In W containing WMoTaTiCr, WTaTiCr, and WTaAlCr alloys, the accelerated oxidation of W and the ensuing WO_3 formation hinder the favorable $CrTaO_4$, $AlTaO_4$, and Al_2O_3 oxide phases to form a continuous protective layer. Hence, a thick oxide layer formed on these alloys. For WMoTaAlCr HEA, the scale layer is mainly consisted of (Al,Cr)TaO_4 oxide with minor amount of WO_3, which provide better protection against oxidation at high temperatures. As observed above, evaporation of Mo oxide occurs during high temperature oxidation of the alloys, which leaves pores and voids in the oxide layers. These volume defects in turn affect the integrity of the protective oxide layer that has a proper PBR value. To increase the oxidation resistance of W containing alloys, the volume mismatch between the oxide layer and non-

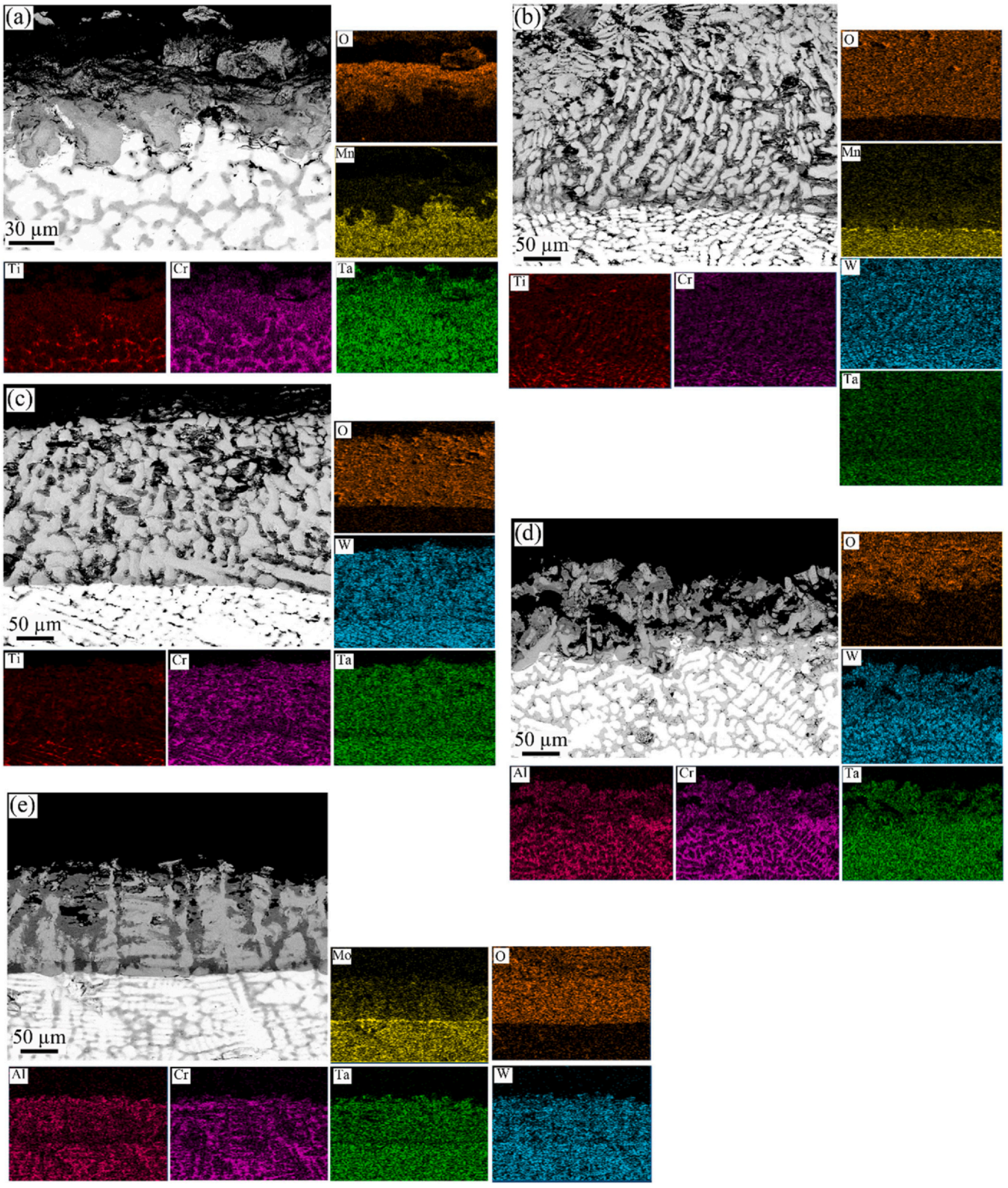


Fig. 6. BSE-SEM micrographs and the corresponding EDS elemental maps of the cross-sections of the sample after 10 h oxidation at 1000 °C for (a) MoTaTiCr, (b) WMoTaTiCr, (c) WTaTiCr, (d) WTaAlCr, (e) WMoTaAlCr HEAs.

oxidized alloy should be compensated by impeding the formation of oxides of too large PBR. It seems to be reasonable that the preferential formation of $CrTaO_4$ and $AlTaO_4$ in W containing alloys could provide such a situation, so that a relatively protective layer quickly forms. However, the preferential oxidation of W should be minimized by removing W-rich dendritic cores through high-temperature annealing or tailoring the microstructure and chemical composition of the alloys.

Table 3
Chemical composition of the oxide layers.

Alloys	EDS chemical composition (at%)	W	Mo	Ta	Ti	Cr	Al	O
MoTaTiCr	Oxide layer	–	1.41	16.50	6.51	6.32	–	69.25
	Alloy	–	23.19	26.23	19.76	19.74	–	11.07
WMoTaTiCr	Oxide layer D	12.20	2.12	9.81	3.64	2.51	–	69.71
	Oxide layer ID	2.70	0.56	7.68	12.28	9.64	–	67.13
	Alloy D	33.14	19.62	23.87	10.21	5.09	–	8.08
	Alloy ID	12.17	18.06	17.26	25.39	23.99	–	3.13
WTaTiCr	Oxide layer D	13.50	–	10.47	4.11	2.49	–	69.43
	Oxide layer ID	7.31	–	8.94	8.07	10.03	–	65.66
	Alloy D	34.02	–	28.22	10.06	5.90	–	21.80
	Alloy ID	3.33	–	18.45	36.47	39.26	–	2.49
WTaAlCr	Oxide layer D	18.01	–	11.51	–	2.29	0.90	67.29
	Oxide layer ID	4.30	–	13.52	–	12.15	6.43	63.38
	Alloy D	46.39	–	31.27	–	7.99	4.93	8.54
	Alloy ID	8.56	–	29.30	–	35.51	23.30	3.34
WMoTaAlCr	Oxide layer D	13.56	3.04	9.63	–	3.02	1.62	69.13
	Oxide layer ID	4.73	4.97	7.88	–	10.64	4.71	67.08
	Alloy D	30.55	22.77	25.79	–	9.34	5.21	6.34
	Alloy ID	5.62	14.56	18.10	–	37.34	23.98	0.40

Table 4
The estimated PBR values for the oxides formed on the surface of the alloys.

Alloy	$CrTaO_4$	$AlTaO_4$	TiO_2	Al_2O_3	WO_3	$Ta_{15}WO_{40.5}$
MoTaTiCr	2.01	–	1.98	–	–	–
WMoTaTiCr	2.00	–	1.97	–	3.20	2.67
WTaTiCr	1.96	–	1.92	–	3.14	2.62
WTaAlCr	1.99	1.95	–	1.34	3.20	2.67
WMoTaAlCr	2.03	1.98	–	1.37	3.26	2.72

4. Conclusion

In summary, we fabricated MoTaTiCr, WMoTaTiCr, WTaTiCr, WTaAlCr, and WMoTaAlCr alloys by arc melting technique. Except WTaAlCr which present some intermetallic compounds, all other as-cast alloys exhibited a single-phase BCC solid solution phase. The quantitative oxidation behavior of the alloys indicated that resistance of W containing alloys against oxidation at 1000 °C was significantly lower than that of MoTaTiCr MEA due to the formation of W oxides, which hinder the formation of a continuous $CrTaO_4$ or $AlTaO_4$ scale layer. It was found that Al addition is a prerequisite to improve the oxidation resistance of W containing HEAs. Owing to the formation of a rutile type $AlTaO_4$ (or $CrTaO_4$) rich scale layer with minor amount of W oxides, WMoTaAlCr HEA exhibited the best resistance against high temperature oxidation among the W containing alloys, but still lower than MoTaTiCr MEA. Although our results indicate a promising prospect to develop oxidation resistant W containing HEAs, further works are required to decrease the content of W oxides and compensate for the Mo losses in the scale layers.

CRediT authorship contribution statement

Ahmad Ostovari Moghaddam: Conceptualization, Methodology, Investigation, Validation, Writing – original draft preparation, Writing – review & editing. **Mikhail Sudarikov**: Investigation, Resources. Nataliya Shaburova: Writing – review & editing. **Dmitry Zherebtsov**: Writing – review & editing. **Vladimir Zhivulin**: Resources. **Ibrokhimi Ashurali Solizoda:** Resources, Investigation. **Andrey Starikov**: Resources. Sergey Veselkov: Resources. **Olga Samoilova**: Writing – review & editing. **Evgeny A. Trofimov**: Conceptualization, Supervision, Resources, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jallcom.2021.162733.

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